

Apprenticeship and Industry Training

Apprenticeship Curriculum Guide

Welder



Apprenticeship
and Industry
Training



Acknowledgements

Apprenticeship and Industry Training (AIT) wishes to express sincere appreciation for the contribution of the many tradespersons, industrial establishments, professional associations, labour organizations, provincial and territorial government departments and agencies, and all others who contributed to this publication.

Special thanks to the following subject matter experts for sharing their knowledge and insights, which were instrumental in developing the standard:

Harold Babuik	Kelly Loren
Natalie Bak	Nicholas Lenko
Colin Bishop	Freddy Liriano
Andrew Connors	Shelley Longman
Richard Dixon	Chris McFee
Peter Hawrylyshen	Josh Post
Xianda Hou	Brad Preece
Terra Kress	Jill Timushka

Additionally, AIT would like to extend its appreciation to the following organizations for their valuable contributions and expert advice throughout the development of the standard:

Alberta Pipe Trades College	Northwestern Polytechnic
Keyano College	Olds College
Lakeland College	Piping Industry Training School
Lethbridge Polytechnic	Portage College
Medicine Hat College	Red Deer Polytechnic
Northern Alberta Institute of Technology	Southern Alberta Institute of Technology
Northern Lakes College	



Program Guide

Description of the Welder Trade	2
Apprenticeship Education Entrance Requirements	3
What To Expect	3
Apprenticeship Pathway to Certification	4
Core Competences	5
Determining Success	5
Thriving	5
Defined Learning	6
Course Content Overview	8
Period One Course Content	13
Period Two Course Content	23
Period Three Course Content	31
Taxonomy Verb List	37

Description of the Welder Trade

Welders in Alberta join and sever metals to fabricate, assemble, repair, and modify structural steel and metal components - such as beams, girders, vessels, and piping - by selecting, setting up, and operating welding, cutting, and gouging equipment; interpreting drawings and welding symbols; preparing and fitting materials; and applying codes, standards, and safety requirements to produce sound welds that meet inspection and quality standards in construction, manufacturing, and industrial maintenance settings.

A journeyperson Welder is an individual who may be able to:

- Apply occupational health and safety requirements.
- Perform safe material handling and access activities.
- Read and interpret drawings, specifications, and welding/NDT symbols.
- Apply trades math.
- Lay out and prepare materials for fabrication.
- Assemble, set up, operate, and maintain oxyfuel equipment.
- Cut and gouge using arc/plasma processes.
- Perform Shielded Metal Arc Welding (SMAW).
- Perform wire-feed welding processes (GMAW/FCAW/MCAW).
- Perform Gas Tungsten Arc Welding (GTAW).
- Apply Submerged Arc Welding (SAW) principles where required.
- Identify metals and apply basic metallurgy concepts.
- Evaluate weld quality and compliance.
- Perform mentorship.
- Use other processes.

Apprenticeship Education Entrance Requirements



Apprenticeship education entrance requirements will **NOT** prevent an individual from registering into an apprentice education program as an apprentice; however, they will restrict that individual's ability to register for the in-classroom training portion of the training until those requirements have satisfied the education entrance requirement for that trade. The minimum education entrance requirements for Welder are:

- English 10-2, Math 10-3
- OR a pass mark in all five Canadian General Educational Development (GED) tests or Canadian Adult Education Credential tests
- OR an Apprenticeship and Industry Training entrance examination.

To increase an individual's chances of success in classroom instruction for the Welder Apprenticeship Education Program, successful completion of English 30-2, Math 30-3, Physics 20 OR Chemistry 20 OR Science 20 and related Career and Technology Studies (CTS) courses are recommended.

What to Expect

Apprenticeship is a training program that combines on-the-job and in-classroom training for a specific profession designated as a trade.

It may be a new form or model of learning to some individuals, however, has been the cornerstone to large portions of Alberta's industry training and economy for almost a century.

In the process of progressing through an apprenticeship education program, the learner experiences both the theoretical and practical components of their chosen discipline and will be assessed in both the classroom and working environments.

The following flow chart presents a visual representation of what a learner can expect while progressing through the Welder program.

Apprentice Pathway to Certification

Apply

Register (Pre-Apprenticeship)

- Choose a program.
- Meet the apprenticeship eligibility requirements.
- Complete the application and include any Prior Learning Assessments



Learn

On-the-job

Welder Period 1

- 1560 hours

Welder Period 2

- 1560 hours

Welder Period 3

- 1560 hours

Apprentices must have an Apprenticeship Education Agreement with a sponsor to earn on-the-job hours.

Classroom Instruction

Welder Period 1

- 240 hours

Welder Period 2

- 240 hours

Welder Period 3

- 240 hours

Apprentices must meet minimum education requirements prior to entrance in classroom instruction.

Competence-based Assessments

Each period, apprentices are assessed:

- on-the-job (by mentors)
- in the classroom (by instructors)
- through period exams (by Apprenticeship and Industry Training)

Earn

Certification

- Post Secondary Education Credential
- Journeyperson Certificate



Core Competences

The foundation of an apprenticeship education program is defined by its core competences. These competences are what the training will focus on as the learner progresses through the Welder Apprenticeship Education Program.

A core competence is an observable skill that is demonstrated by the learner. Competences are identified by industry subject matter experts and cover the job-related tasks and activities that a Welder will be required to perform in the course of their employment.

Determining Success

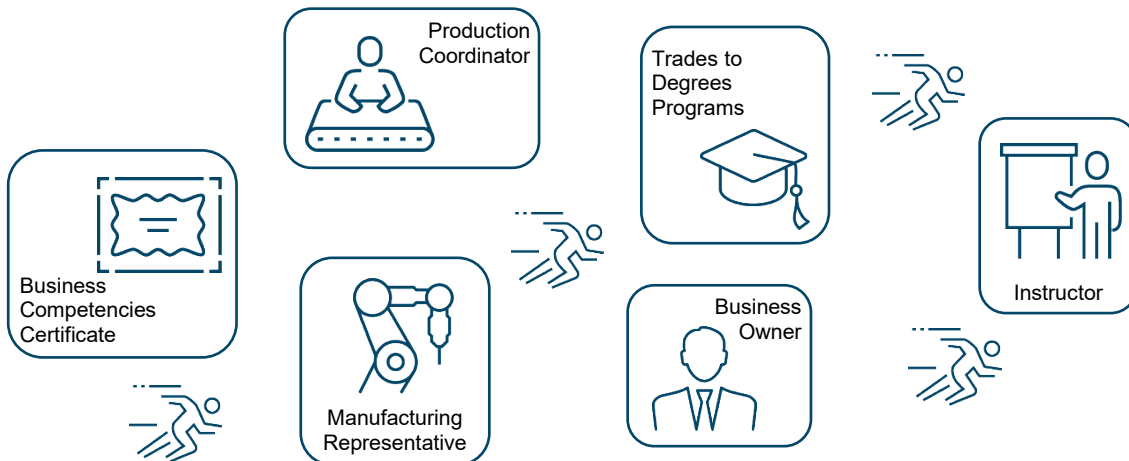
Grades for the theory component of technical training (classroom instruction) are calculated and based on a cumulative weighted average. To complete a period of classroom instruction successfully, an apprentice must achieve a minimum cumulative weighted average of 65% across all sections in the classroom instruction component of their apprenticeship education program with no grades below 50% in any one section.

Lab and shop components are calculated as a cumulative weighted average requiring a minimum of 65% with a minimum grade of 65% in each section.

Thriving

Successful completion of the Welder Apprenticeship Education Program is an accomplishment that should be celebrated and can continue to be a successful provider and career choice for a lifetime.

It can also become a useful lever that opens doors leading to future career choice enhancers and/or create alternate professions pathways. This may include some of the following:



Defined Learning



This curriculum guide was developed based on the core competences for this trade.

The Core Competence Statements document can be accessed through this QR code or found by visiting the Alberta Apprenticeship and Industry Training (AIT) website (tradesecrets.alberta.ca)



Section

Each section in a period supports a core competence for this trade. The section is the highest-level division in a curriculum guide that groups related topics under one title.

Weighting

Each section is assigned a weighting (in percent and hours) by industry to reflect the approximate amount of available time that is dedicated to it, on average, in the learning environment.

Example: Period One Course Content

Section Title	Weighting	
	Hours	%
Foundational Skills, Job Responsibilities and Procedures	65	27%
Tools, Equipment, and Instruments	22	9%
Drawings and Specifications	24	10%
Rigging: Hoisting, Lifting and Load-Moving	67	28%
Cutting and Fabrication	62	26%
TOTAL	240	100%

Topic

A topic is an area of focus within a section that identifies an outcome and its associated objectives. Each topic has its own assigned weighting in hours and percentages.

Outcome

An outcome states what a student will be able to do within an area of focus.

Objective

An objective is a specific, measurable step that a learner must do to achieve the topic's outcome.

Taxonomy

Taxonomies are a cognitive framework of learning behaviors organized hierarchically in categories: knowledge, comprehension, application, analysis, evaluation, and synthesis. Taxonomy is used as a tool to identify and create learning objectives that define and measure the learning experience for all stakeholders involved.

For the purposes of Alberta's apprenticeship education assessments, taxonomies are classified into three of these categories:

- Taxonomy I competences focus on recollection of facts, definitions, and principles.
- Taxonomy II competences are a procedural application of knowledge (i.e. reference to a step-by-step process).
- Taxonomy III competences deal with critical thinking and problem solving (i.e. diagnosis and analysis).

These categories identify the type of skill set that will be applied to be successful.

Some topics have multiple taxonomies that will require the training to address different skill sets.



NOTE: A taxonomy verb list is also attached in the appendix to demonstrate the level and/or complexity of a given taxonomy.

Course Content Overview

PERIOD ONE COURSE CONTENT

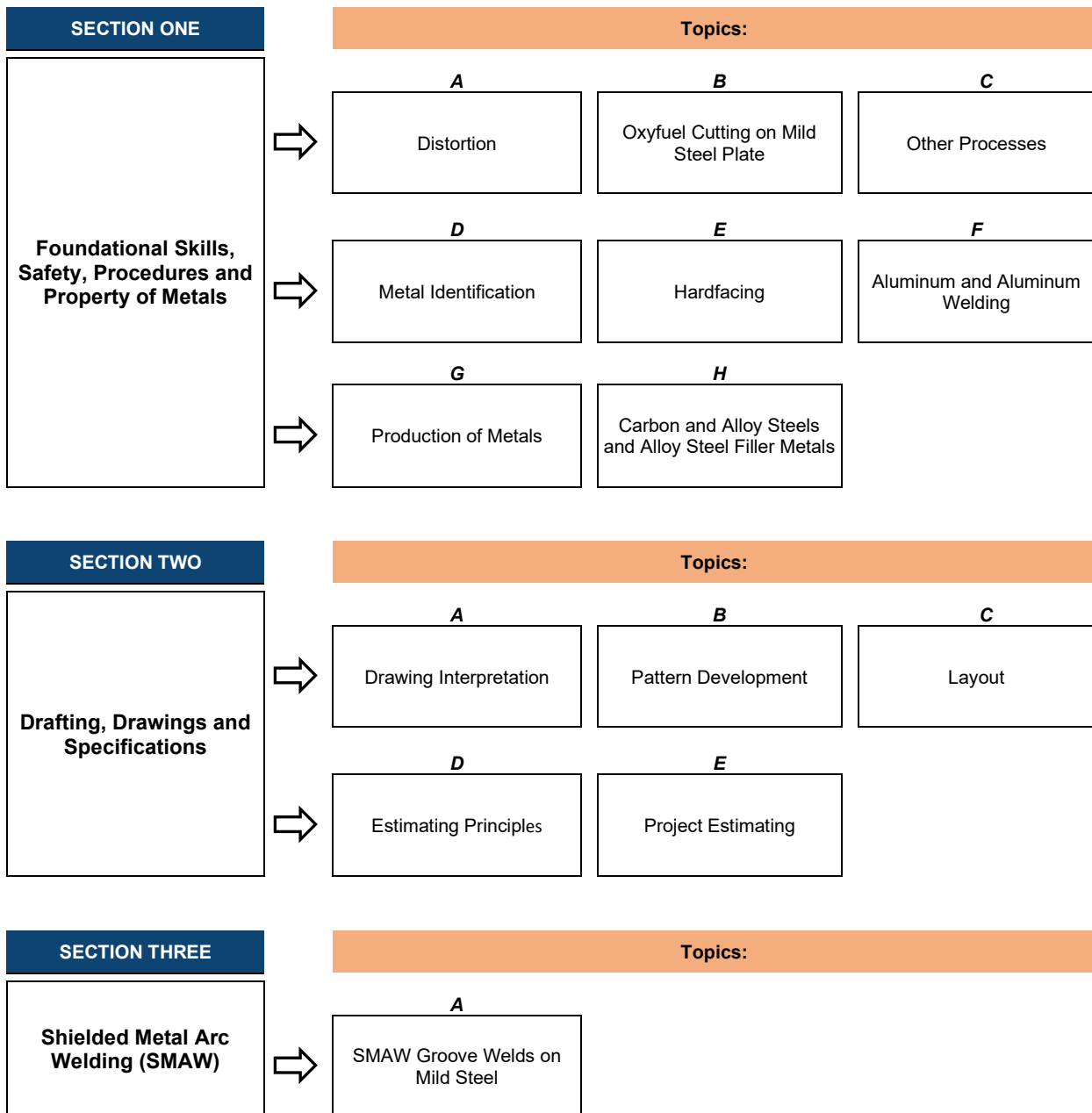
SECTION ONE		Topics:		
Foundational Skills, Safety and Procedures	⇒	A Welder Apprenticeship Training Program Orientation	B Safety, Legislation, Regulations & Industry Policy in the Trades	C Climbing and Material Handling: Lifting, Rigging and Hoisting
	⇒	D Hazardous Materials and Fire Protection	E Welding Safety	F Oxyfuel Equipment and Cutting
	⇒	G Weld Faults	H Hand Tools	I Power Tools
	⇒	J Arc Cutting and Gouging		
SECTION TWO		Topics:		
Drafting, Drawings and Specifications	⇒	A Welding Symbols	B Fractions	C Decimals
	⇒	D Ratios, Percentage and Measurement Systems	E Geometric Formulas	F Joint and Weld Types
SECTION THREE		Topics:		
Shielded Metal Arc Welding (SMAW)	⇒	A SMAW Equipment	B Mild Steel Electrodes	C SMAW Fillet Welds on Mild Steel

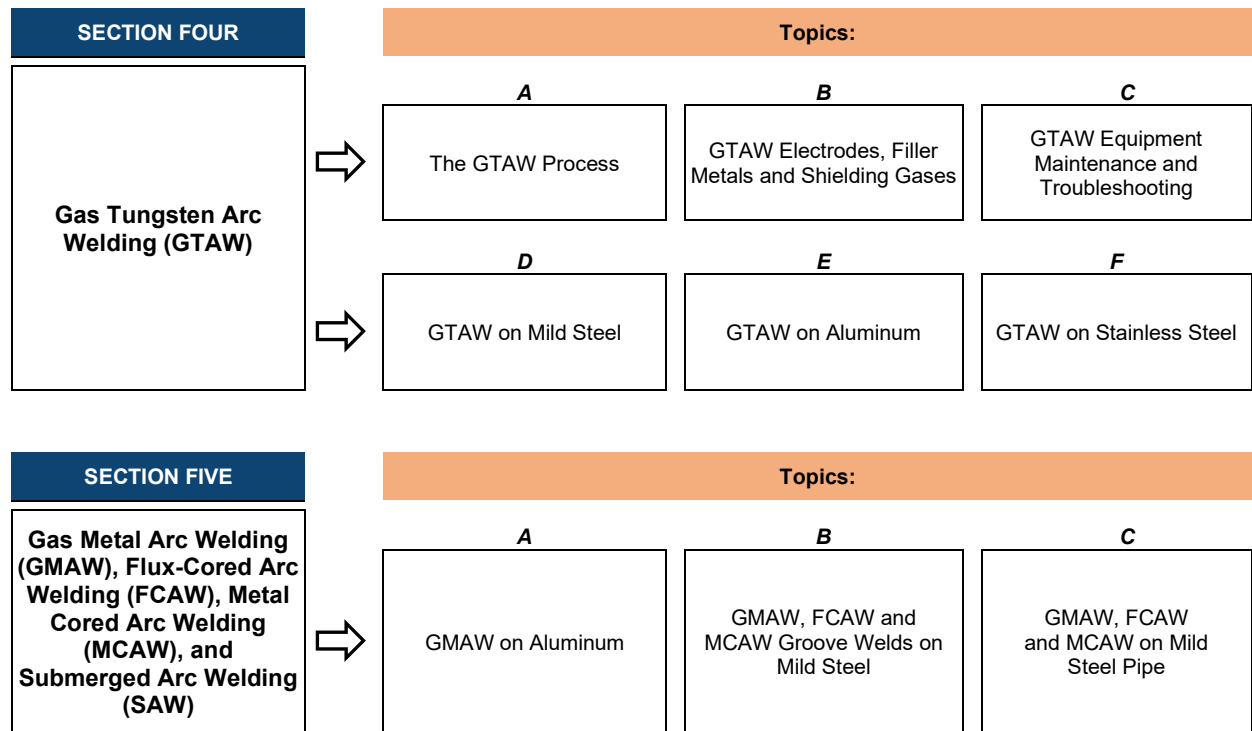
SECTION FOUR

Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), Metal Cored Arc Welding (MCAW), and Submerged Arc Welding (SAW)

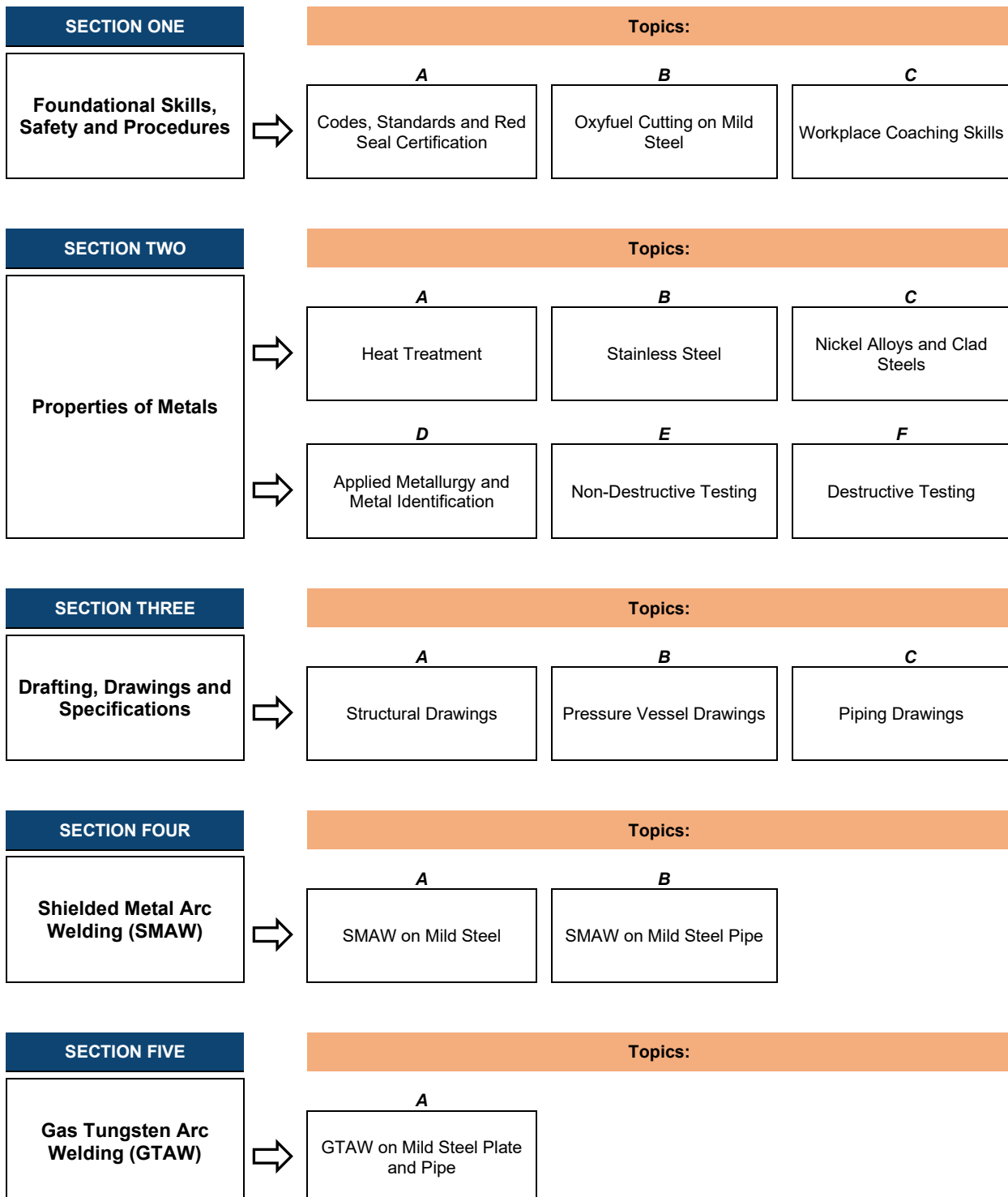
Topics:		
A	B	C
Wire Feed Welding Equipment Power Sources	Wire Feed Welding Filler Metals and Feeders	Wire Feed Welding Shielding Gases
D	E	F
Wire Welding Maintenance and Troubleshooting	Submerged Arc Welding (SAW)	GMAW on Mild Steel
G		
FCAW and MCAW on Mild Steel		

PERIOD TWO COURSE CONTENT





PERIOD THREE COURSE CONTENT



Period One Course Content

(8 weeks – 240 hours)

Section Title		Weighting	
		Hours	%
Section One:	Foundational Skills, Safety and Procedures	70 hrs	29%
Section Two:	Drafting, Drawings and Specifications	40 hrs	17%
Section Three:	Shielded Metal Arc Welding (SMAW)	43 hrs	18%
Section Four:	Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), Metal Cored Arc Welding (MCAW), and Submerged Arc Welding (SAW)	87 hrs	36%
TOTAL		240 hrs	100%

Section One: Foundational Skills, Safety and Procedures



This section establishes entry-level readiness for welding work by covering apprenticeship orientation, OHS safety and regulations, rigging/material handling, and hazardous materials/fire protection, then building hands-on skills in oxyfuel equipment setup and cutting, tool use, arc cutting/gouging, and weld fault identification.

Topic A.	Welder Apprenticeship Training Program Orientation	Weighting	
		3 Hours	1%

Outcome: Upon successful completion, learners will be able to explain how welder apprenticeship training works in Alberta, including curriculum and attendance expectations, contract and portfolio responsibilities, employment sectors, and external organizations affecting the trade.

- Objectives:**
1. Explain the Welder curriculum guide learning outcomes and objectives.
 2. Describe the responsibilities for the contract of apprenticeship and competency portfolio by the apprentice, sponsor and Alberta Apprenticeship and Industry Training.
 3. Identify industrial, commercial and construction fields that provide employment opportunities for welders.
 4. Define the role of external organizations that affect the welding trade.

Topic B.	Safety, Legislation, Regulations and Industry Policy in the Trades	Weighting	
		3 Hours	1%

Outcome: Upon successful completion, learners will be able to apply the legislation, regulations, and industry practices required for safe work in this trade by completing hazard assessment and control, following emergency procedures, and selecting, maintaining, and using required PPE in accordance with OHS and related requirements.

- Objectives:**
1. Demonstrate the application of the Occupational Health and Safety (OHS) Act, Regulation and Code.
 2. Describe the sponsor's and employee's role with OHS regulations, Worksite Hazardous Materials Information Systems (WHMIS), fire regulations, Workers Compensation Board (WCB) regulations and related advisory bodies and agencies.
 3. Describe and comply with industry practices for hazard assessment and control procedures.
 4. Describe the responsibilities of worker and sponsors to apply emergency procedures.
 5. Describe tradesperson attitudes with respect to housekeeping, personal protective equipment (PPE) and emergency procedures.
 6. Describe the roles and responsibilities of sponsors and employees with the selection and use of PPE.
 7. Maintain required PPE for tasks.
 8. Use required PPE for tasks.

Topic C.	Climbing and Material Handling: Lifting, Rigging and Hoisting	Weighting	
		6 Hours	3%

Outcome: Upon successful completion, learners will be able to apply correct climbing, lifting, rigging, and hoisting practices by following material handling and storage procedures, calculating load weight and centre of gravity, verifying sling and rope capacity limits, and using lifting and rigging equipment.

- Objectives:**
1. Climbing, lifting, rigging and hoisting safety.
 2. Identify procedures for handling and storing materials.
 3. Determine weight and centre of gravity of loads.
 4. Identify the load limits of wire rope and synthetic slings.
 5. Describe the use of plate clamps and cable clips.

Topic D.	Hazardous Materials and Fire Protection	Weighting	
		2 Hours	1%

Outcome: Upon successful completion, learners will be able to safely apply hazardous materials and fire protection practices by using handling, storage, transport, and venting procedures, and by identifying fire hazards/classes and using appropriate fire protection equipment.

- Objectives:**
1. Describe handling, storing and transporting procedures for hazardous material.
 2. Describe venting procedures when working with hazardous materials.
 3. Describe hazards, classes, procedures and equipment related to fire protection.

Topic E.	Welding Safety	Weighting	
		3 Hours	1%

Outcome: Upon successful completion, learners will be able to apply safe work practices during welding and cutting operations to identify hazards, select and use PPE, apply controls for welding fumes and gases, use electrical safety precautions, and follow procedures for work in confined spaces or potentially dangerous enclosures.

- Objectives:**
1. Identify hazards for welding and cutting operations.
 2. Identify the use of personal protective equipment for welding and cutting operations.
 3. Explain the hazards and controls involved with welding fumes and gases.
 4. Explain the effects of electricity and precautions used to prevent injury.
 5. Describe the procedure for welding or cutting in confined spaces or potentially dangerous enclosures.

Topic F.	Oxyfuel Equipment and Cutting	Weighting	
		37 Hours	15%

Outcome: Upon successful completion, learners will be able to assemble, adjust, operate, shut down, and maintain oxyfuel equipment; perform hand-held and machine oxyfuel cuts on mild steel plate, pipe, and structural shapes.

- Objectives:**
1. Identify causes and preventive measures for backfires, flashbacks and burn backs.
 2. Describe how to operate a hand-held oxyfuel cutting torch on mild steel plate and structural shapes.
 3. Describe the characteristics and handling procedures for oxygen and fuel gases.
 4. Describe the functions of oxyfuel equipment components.
 5. Explain the procedure for placement, set-up and shutting down of oxyfuel equipment.
 6. Demonstrate the use, care and maintenance of oxyfuel equipment components.
 7. Describe pressure and flame adjustments.
 8. Perform straight line, bevel, and shape cutting on mild steel.
 9. Pierce and cut holes in mild steel plate.
 10. Cope 3/8 in. mild steel to fit a 100 mm (4 in.) C shape.
 11. Perform cuts on structural shapes.
 12. Operate a machine oxyfuel cutting torch on mild steel plate and pipe.

Topic G.	Weld Faults	Weighting	
		4 Hours	2%

Outcome: Upon successful completion, learners will be able to define classifications of weld faults, identify weld faults, their causes, and methods of prevention.

- Objectives:**
1. Define the classifications of weld faults.
 2. Identify weld faults, their causes and methods of prevention.

Topic H.	Hand Tools	Weighting	
		3 Hours	1%

Outcome: Upon successful completion, learners will be able to describe required safety precautions and identify layout, measuring, clamping, cutting, and other hand tools.

- Objectives:**
1. Describe safety precautions for hand tools.
 2. Identify the layout and measuring tools and their uses.
 3. Identify clamping tools and their uses.
 4. Identify cutting tools and their uses.
 5. Identify the other hand tools used by welders.

Topic I.	Power Tools	Weighting	
		3 Hours	1%

Outcome: Upon successful completion, learners will be able to demonstrate the operation of grinders, describe the operation of drills, metal forming/shaping, metal cutting tools, and the use of power positioners.

- Objectives:**
1. Demonstrate the operation of grinders.
 2. Describe the operation of drills.
 3. Describe the operation of metal forming and shaping tools.
 4. Describe the operation for metal cutting tools.
 5. Describe the use of power positioners.

Topic J.	Arc Cutting and Gouging	Weighting	
		6 Hours	3%

Outcome: Upon successful completion, learners will be able to apply safe work practices during cutting operations to identify hazards, select and use PPE.

- Objectives:**
1. Describe the plasma arc cutting process and equipment.
 2. Demonstrate plasma arc cutting.
 3. Describe the carbon arc cutting process and equipment.
 4. Demonstrate carbon arc cutting.

Section Two: Drafting, Drawings and Specifications



This section covers weld and Non-Destructive Testing (NDT) symbols, essential trades math and measurement, and basic joint and weld types for interpreting fabrication requirements.

Topic A.	Weld Symbols	Weighting	
		6 Hours	2%

Outcome: Upon successful completion, learners will be able to explain the purpose of welding symbols, define and interpret weld, welding, and supplementary symbols; interpret NDT symbols.

- Objectives:**
1. Explain the purpose of welding symbols.
 2. Define weld, welding, and supplementary symbols.
 3. Interpret weld, welding, and supplementary symbols.
 4. Identify the dimensioning of welding symbols.
 5. Interpret NDT symbols.

Topic B.	Fractions	Weighting	
		8 Hours	3%

Outcome: Upon successful completion, learners will be able to identify key fraction terms and concepts, use practical fractions on a tape measure, and solve fraction problems.

- Objectives:**
1. Identify terms and concepts used with fractions.
 2. Use practical fractions on a tape measure.
 3. Solve problems using fractions.

Topic C.	Decimals	Weighting	
		4 Hours	2%

Outcome: Upon successful completion, learners will be able to convert fractions to decimals, and decimals to fractions to solve practical problems.

- Objectives:**
1. Round decimal fractions to specified place values.
 2. Add, subtract, multiply and divide decimal fractions.
 3. Convert fractions to decimals.
 4. Convert decimal inches and decimal feet, to feet and inch fractions with a practical denominator.
 5. Solve decimal fraction calculations.

Topic D.	Ratios, Percentage and Measurement Systems	Weighting	
		6 Hours	3%

Outcome: Upon successful completion, learners will be able to solve problems using ratios, proportions, and percentages; identify and convert between fractions, decimals, percentages, and units of measure.

- Objectives:**
1. Convert between fractions, decimals and percentages.
 2. Solve percentage problems.
 3. Calculate ratio problems: two quantities in the form of a ratio and two ratios in the form of a proportion.
 4. Identify metric units of measure.
 5. Convert imperial units: feet to inches, square inches to square feet, and cubic measures to gallons.
 6. Convert between units of measure.

Topic E.	Geometric Formulas	Weighting	
		12 Hours	5%

Outcome: Upon successful completion, learners will be able to apply geometric formulas and perform calculations to solve problems.

- Objectives:**
1. Identify formulas and solve problems for perimeter, area and volume.
 2. Calculate the weight of a solid.
 3. Calculate the capacity of a container in gallons and liters.

Topic F.	Joint and Weld Types	Weighting	
		4 Hours	2%

Outcome: Upon successful completion, learners will be able to identify joint and weld types; describe weld dimensions and variations.

- Objectives:**
1. Identify the five basic joints.
 2. Describe the types of welds and their dimensions.
 3. Identify joint and weld type variations.

Section Three: Shielded Metal Arc Welding (SMAW)



This section covers SMAW equipment and electrodes, then builds hands-on skills to produce welds on mild steel in the required positions.

Topic A.	SMAW Equipment	Weighting	
		8 Hours	3%

Outcome: Upon successful completion, learners will be able to identify SMAW equipment and power sources and explain key concepts.

- Objectives:**
1. Define SMAW related terms.
 2. Identify welding cables and accessories for welding power sources.
 3. Identify the effect of arc length on amperage and voltage.
 4. Describe AC and AC-DC rectified power sources.
 5. Describe AC and DC generator power sources.
 6. Describe multi-process inverter power sources.

Topic B.	Mild Steel Electrodes	Weighting	
		5 Hours	2%

Outcome: Upon successful completion, learners will be able to identify and select mild steel SMAW electrodes.

- Objectives:**
1. Define the terms associated with SMAW electrodes.
 2. Identify the CSA and AWS classification and specifications for SMAW electrodes.
 3. Identify the types and functions of SMAW electrode coatings.
 4. Describe the functions of the slag.
 5. Describe care, handling and storage procedures for these electrodes.
 6. Identify mild steel SMAW electrodes and their applications.

Topic C.	SMAW Fillet Welds on Mild Steel	Weighting	
		30 Hours	13%

Outcome: Upon successful completion, learners will be able to perform SMAW welds on mild steel in the required positions.

- Objectives:**
1. Perform surface welds in the flat position using E4310 (E6010) and E4918 (E7018) electrodes.
 2. Perform fillet welds in the 1F, 2F, 3F and 4F positions using E4310 (E6010) and E4918 (E7018) electrodes.

Section Four: Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), Metal Cored Arc Welding (MCAW), and Submerged Arc Welding (SAW)



This section covers wire-feed welding setup and troubleshooting and builds hands-on skills to perform GMAW/FCAW/MCAW welds on mild steel in required positions.

Topic A.	Wire Feed Welding Equipment Power Sources	Weighting	
		6 Hours	3%

Outcome: Upon successful completion, learners will be able to identify and describe wire feed welding equipment.

- Objectives:**
1. Describe the principles of operation of wire feed welding equipment.
 2. Identify the components of a wire feed welding equipment set-up.
 3. Describe wire feed welding equipment power sources and wire feeders.
 4. Identify advantages and disadvantages of wire feed processes.

Topic B.	Wire Feed Welding Filler Metals and Feeders	Weighting	
		6 Hours	3%

Outcome: Upon successful completion, learners will be able to identify and describe wire feed consumables and feeder components.

- Objectives:**
1. Identify wire feed welding equipment filler metals.
 2. Describe the modes of metal transfer.
 3. Describe wire feed drive systems and gun and cable accessories.
 4. Describe wire feed operating variables.

Topic C.	Wire Feed Welding Shielding Gases	Weighting	
		6 Hours	3%

Outcome: Upon successful completion, learners will be able to identify shielding gases and supply systems for wire feed processes.

- Objectives:**
1. Identify shielding gases for wire feed processes.
 2. Identify shielding gas supply systems.

Topic D.	Wire Feed Welding Maintenance and Troubleshooting	Weighting	
		6 Hours	2%

Outcome: Upon successful completion, learners will be able to set up, maintain, and troubleshoot wire feed welding equipment.

- Objectives:**
1. Demonstrate the set-up and maintenance required for wire feed drive systems and gun assemblies.
 2. Perform corrective measures for malfunctioning wire process equipment.
 3. Describe the effects of angle and inclination.

Topic E.	Submerged Arc Welding (SAW)	Weighting	
		3 Hours	1%

Outcome: Upon successful completion, learners will be able to identify and describe the components, consumables, and operation of the SAW process.

- Objectives:**
1. Describe the operating principles of SAW.
 2. Identify the components of SAW.
 3. Describe SAW power sources and equipment.
 4. Describe SAW operating variables.
 5. Identify SAW filler metals and fluxes.
 6. Identify advantages and disadvantages of SAW.

Topic F.	GMAW on Mild Steel	Weighting	
		30 Hours	12%

Outcome: Upon successful completion, learners will be able to perform GMAW welds on mild steel.

- Objectives:**
1. Perform stringer and weave beads in the flat and horizontal positions.
 2. Perform welds in 1F, 2F, and 3F positions.
 3. Perform welds in 1G, 2G, 3G positions.

Topic G.	FCAW and MCAW on Mild Steel	Weighting	
		30 Hours	12%

Outcome: Upon successful completion, learners will be able to perform FCAW and demonstrate MCAW welding on mild steel.

Topic G.	FCAW and MCAW on Mild Steel	Weighting	
		30 Hours	12%

- Objectives:**
1. Perform stringer and weave beads in the flat and horizontal positions using FCAW on mild steel plate.
 2. Perform welds in 1F, 2F and 3F positions using FCAW process.
 3. Demonstrate MCAW process.
 4. Perform welds in 1GF, 2G, 3GF and 4GF joint configurations using FCAW process with a backing plate.

Period Two Course Content

(8 weeks – 240 hours)

Section Title		Weighting	
		Hours	%
Section One:	Foundational Skills, Safety, Procedures and Properties of Metals	39 hrs	16%
Section Two:	Drafting, Drawings and Specifications	51 hrs	21%
Section Three:	Shielded Metal Arc Welding (SMAW)	40 hrs	17%
Section Four:	Gas Tungsten Arc Welding (GTAW)	56 hrs	23%
Section Five:	Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), Metal Cored Arc Welding (MCAW), and Submerged Arc Welding (SAW)	54 hrs	23%
TOTAL		240 hrs	100%

Section One: Foundational Skills, Safety, Procedures and Properties of Metals



This section covers welding skills, metal properties, and an overview of other processes.

Topic A.	Distortion	Weighting	
		2 Hours	1%

Outcome: Upon successful completion, learners will be able to identify the types, causes, and control methods of welding distortion.

Objectives:

1. Identify how heat and temperature relate to distortion.
2. Identify the three types of distortion, their causes and control of each type.

Topic B.	Oxyfuel Cutting on Mild Steel Plate	Weighting	
		20 Hours	8%

Outcome: Upon successful completion, learners will be able to perform oxyfuel cutting on mild steel plate.

- Objectives:**
1. Perform piercing and cutting of an elliptical opening.
 2. Perform a cope on 10 mm (3/8 in.) mild steel to fit a 100 mm (4 in.) C shape and complete a 30° bevel.

Topic C.	Other Processes	Weighting	
		2 Hours	1%

Outcome: Upon successful completion, learners will be able to describe other welding-related processes and their applications.

- Objectives:**
1. Describe robotics, handheld laser beam welding, resistance spot welding, thermite, friction stir welding, and 3D metal printing.

Topic D.	Metal Identification	Weighting	
		3 Hours	1%

Outcome: Upon successful completion, learners will be able to identify and describe metals by their physical and mechanical properties.

- Objectives:**
1. Identify metals by physical and mechanical properties.
 2. Describe chip, spark, file hardness and flame tests.
 3. Describe the mechanical properties of metals.
 4. Describe the physical properties of metals.

Topic E.	Hardfacing	Weighting	
		2 Hours	1%

Outcome: Upon successful completion, learners will be able to describe and identify hardfacing applications, procedures, and consumables.

- Objectives:**
1. Describe the hardfacing process and applications.
 2. Identify the types of wear.
 3. Identify filler metals for hardfacing.
 4. Identify the problems associated with hardfacing and how to avoid them.
 5. Describe the procedures for applying hardfacing materials with filler wires.

Topic F.	Aluminum and Aluminum Welding	Weighting	
		3 Hours	1%

Outcome: Upon successful completion, learners will be able to describe aluminum properties and welding principles, including alloy designations.

- Objectives:**
1. Describe the physical and chemical properties of aluminum versus steel.
 2. Describe how physical and chemical properties affect the welding of aluminum.
 3. Describe the Aluminum Association numerical designation for casting alloys and wrought aluminum.
 4. Describe the effects of welding on heat treatable and non-heat treatable alloys.
 5. Identify the filler metals used for welding aluminum.

Topic G.	Production of Metals	Weighting	
		3 Hours	1%

Outcome: Upon successful completion, learners will be able to describe the production processes and types of iron and steel.

- Objectives:**
1. Describe the production processes for iron and steel.
 2. Describe the types of iron and steel.

Topic H.	Carbon and Alloy Steels and Alloy Steel Filler Metals	Weighting	
		4 Hours	2%

Outcome: Upon successful completion, learners will be able to identify and describe carbon and alloy steels and their properties.

- Objectives:**
1. Identify the carbon content and the uses for low, medium and high carbon steel.
 2. Describe the effects of carbon content on the weldability of steel.
 3. Identify the effects of elements in the properties of carbon steel.
 4. Identify the alloying elements in alloy steels.
 5. Identify the types, properties and weldability of low alloy steels.
 6. Identify the properties and weldability of high strength low alloy steels.
 7. Identify alloy steel filler material classifications using AWS and CSA specifications.
 8. Identify low alloy steel filler metals and their applications.

Section Two: Drafting, Drawings and Specifications



This section builds skills to create and interpret drawings, apply layout methods, and complete job estimates.

Topic A.	Drawing Interpretation	Weighting	
		2 Hours	1%

Outcome: Upon successful completion, learners will be able to interpret drawings.

- Objectives:**
1. Identify the alphabet of lines.
 2. Describe the purpose of drawings.
 3. Identify elements and information found on drawings.
 4. Interpret symbols, views and sections used on drawings.
 5. Identify metric and imperial dimensioning.

Topic B.	Pattern Development	Weighting	
		22 Hours	8%

Outcome: Upon successful completion, learners will be able to identify and describe drawing tools and methods for pattern development; perform layouts.

- Objectives:**
1. Describe the principles of scale drawings.
 2. Describe the principles of perspective, oblique and isometric drawings.
 3. Describe and sketch orthographic projection.
 4. Develop an orthographic drawing to scale.
 5. Describe drawing tools.
 6. Describe the parts of geometric shapes and angles.
 7. Perform layouts.

Topic C.	Layout	Weighting	
		12 Hours	5%

Outcome: Upon successful completion, learners will be able to identify, describe, and perform layout procedures.

- Objectives:**
1. Describe layout abbreviations and symbols.
 2. Describe layout tools and mark-up methods.
 3. Describe templates.
 4. Describe the procedure for plate utilization.
 5. Identify pipe sizes and schedules.

Topic C.	Layout	Weighting	
		12 Hours	5%
	6. Describe pipe layout tools.		
	7. Perform layout for plate and pipe.		

Topic D.	Estimating Principles	Weighting	
		6 Hours	3%

Outcome: Upon successful completion, learners will be able to demonstrate skills related to estimating.

- Objectives:**
1. Convert angular measurements to linear dimensions.
 2. Calculate the cost of steel.

Topic E.	Project Estimating	Weighting	
		9 Hours	4%

Outcome: Upon successful completion, learners will be able to develop a project estimate.

- Objectives:**
1. Determine total costs for a project.
 2. Develop a project estimate.

Section Three: Shielded Metal Arc Welding (SMAW)



This section develops hands-on skills in SMAW groove welding on mild steel.

Topic A.	SMAW Groove Welds on Mild Steel	Weighting	
		40 Hours	17%

Outcome: Upon successful completion, learners will be able to perform SMAW groove welds on mild steel.

- Objectives:**
1. Perform welds in the 1G, 2G, 3G using E4310 (E6010) root and E4918 (E7018) fill and cap.
 2. Perform welds in the 1GF, 2G, 3GF using E4918 (E7018) with backing plate.

Section Four: Gas Tungsten Arc Welding (GTAW)



This section introduces GTAW fundamentals, equipment, consumables, and troubleshooting; develop hands-on skills.

Topic A.	The GTAW Process	Weighting	
		5 Hours	2%

Outcome: Upon successful completion, learners will be able to identify and describe GTAW procedures and fundamentals.

- Objectives:**
1. Describe the GTAW process and applications.
 2. Describe advantages and disadvantages of the GTAW process.
 3. Identify hazards and protective measures associated with GTAW.
 4. Describe single-phase and three-phase power.
 5. Identify components of a GTAW workstation.
 6. Describe types of GTAW power sources.
 7. Identify AC, DC, and high frequency welding currents used in GTAW.
 8. Describe torch assemblies.
 9. Describe gas regulators and flow meters.

Topic B.	GTAW Electrodes, Filler Metals and Shielding Gases	Weighting	
		7 Hours	3%

Outcome: Upon successful completion, learners will be able to identify and describe GTAW electrodes, filler metals, and shielding gases.

- Objectives:**
1. Identify electrode designations and explain their applications.
 2. Identify the function of the filler metals in GTAW.
 3. Identify filler metals by AWS and CSA designations and explain their applications.
 4. Describe the care and preparation of electrodes and filler metals.
 5. Identify the function of the electrode in GTAW.
 6. Describe the types and applications of shielding gases used in GTAW.
 7. Describe the advantages and disadvantages of various shielding gases.

Topic C.	GTAW Equipment Maintenance and Troubleshooting	Weighting	
		4 Hours	2%

Outcome: Upon successful completion, learners will be able to troubleshoot and maintain GTAW equipment.

Topic C.	GTAW Equipment Maintenance and Troubleshooting	Weighting	
		4 Hours	2%

- Objectives:**
1. Describe the set up and maintenance of GTAW equipment.
 2. Determine and solve power source output current problems.
 3. Determine and solve GTAW torch and cable assembly problems.
 4. Determine and solve shielding gas coverage problems.

Topic D.	GTAW on Mild Steel	Weighting	
		30 Hours	12%

Outcome: Upon successful completion, learners will be able to perform GTAW on mild steel.

- Objectives:**
1. Perform touch start, lift start and high frequency methods.
 2. Perform stringer beads in the flat position on mild steel gauge plate.
 3. Perform joint preparation for GTAW on mild steel gauge plate.
 4. Perform 1F, 2F, 3F welds on mild steel gauge plate.
 5. Perform 1G, 2G, 3G welds on plate and 2G, 5G welds on pipe.

Topic E.	GTAW on Aluminum	Weighting	
		5 Hours	2%

Outcome: Upon successful completion, learners will be able to perform GTAW on aluminum.

- Objectives:**
1. Perform stringer beads in the flat position on aluminum gauge plate.
 2. Perform 1F, 2F, 3F welds on aluminum gauge plate.

Topic F.	GTAW on Stainless Steel	Weighting	
		5 Hours	2%

Outcome: Upon successful completion, learners will be able to perform GTAW on stainless steel.

- Objectives:**
1. Perform 2F, 3F welds on stainless steel gauge plate.

Section Five: Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), Metal Cored Arc Welding (MCAW), and Submerged Arc Welding (SAW)



This section introduces GMAW, FCAW, MCAW with hands-on welding on aluminum and mild steel plate and pipe.

Topic A.	GMAW on Aluminum	Weighting	
		2 Hours	1%

Outcome: Upon successful completion, learners will be able to perform GMAW on aluminum.

Objectives:

1. Perform stringer and weave beads in the flat and horizontal positions.
2. Perform 1F, 2F, 3F welds.

Topic B.	GMAW, FCAW and MCAW Groove Welds on Mild Steel	Weighting	
		40 Hours	17%

Outcome: Upon successful completion, learners will be able to perform GMAW, FCAW, and MCAW groove welds on mild steel.

Objectives:

1. Perform 1G, 2G, 3G welds on mild steel using GMAW root, and FCAW or MCAW fill and cap.
2. Demonstrate welds with MCAW.
3. Perform 3G GMAW root and 4G fill and cap with FCAW.

Topic C.	GMAW, FCAW and MCAW on Mild Steel Pipe	Weighting	
		12 Hours	5%

Outcome: Upon successful completion, learners will be able to perform GMAW, FCAW, MCAW pipe welding.

Objectives:

1. Perform 2G weld on pipe using GMAW.
2. Perform 1GR weld on pipe using a GMAW root pass and FCAW or MCAW fill and cap.
3. Perform 2G weld on pipe using GMAW root pass and FCAW fill and cap.
4. Perform 5G weld on pipe using a downhill root pass, and an uphill fill and cap.

Period Three Course Content

(8 weeks – 240 hours)

Section Title		Weighting	
		Hours	%
Section One:	Foundational Skills, Safety and Procedures	44 hrs	18%
Section Two:	Properties of Metals	26 hrs	11%
Section Three:	Drafting, Drawings and Specifications	48 hrs	20%
Section Four:	Shielded Metal Arc Welding (SMAW)	82 hrs	34%
Section Five:	Gas Tungsten Arc Welding (GTAW)	40 hrs	17%
TOTAL		240 hrs	100%

Section One: Foundational Skills, Safety and Procedures



This section builds core readiness for welder training in the workplace, including codes, standards, and Red Seal exam preparation; develop oxyfuel cutting skills on mild steel, and introduce workplace coaching practices.

Topic A.	Codes, Standards and Red Seal Certification	Weighting	
		12 Hours	5%

Outcome: Upon successful completion, learners will be able to identify and describe welding codes and standards and use Red Seal resources to prepare for the Interprovincial exam.

- Objectives:**
1. Define the terms code, standard and specification.
 2. Identify the advantages of standardization.
 3. Identify agencies that set codes and standards.
 4. Identify the codes that govern welding.
 5. Describe the welding procedure qualification.
 6. Describe the welder performance qualification.
 7. Use the Red Seal Occupational Standard (RSOS) to prepare for an Interprovincial examination.

Topic B.	Oxyfuel Cutting on Mild Steel	Weighting	
		30 Hours	13%

Outcome: Upon successful completion, learners will be able to perform oxyfuel cutting procedures on mild steel.

Objectives:

1. Perform oxyfuel bevel cutting on mild steel.
2. Perform piercing, cutting, and coping on mild steel.

Topic C.	Workplace Coaching Skills	Weighting	
		2 Hours	0%

Outcome: Upon successful completion, learners will be able to describe and apply the workplace coaching process.

Objectives:

1. Describe the process and skills for coaching an apprentice, including setting expectations, demonstrating tasks, observing performance, and providing feedback.

Section Two: Properties of Metals



This section provides an overview of metal properties, metallurgy, and testing methods.

Topic A.	Heat Treatment	Weighting	
		4 Hours	2%

Outcome: Upon successful completion, learners will be able to define and describe heat treatment fundamentals.

Objectives:

2. Define heat-affected zones in metals.
3. Describe the difference between heat and temperature.
4. Describe the three forms of heat transfer.
5. Describe the effects of expansion and contraction.
6. Describe the purpose and effects of preheat and postheat.
7. Describe the practices of heat treatment.
8. Describe the principle of temperature-indicating devices.

Topic B.	Stainless Steel	Weighting	
		4 Hours	2%

Outcome: Upon successful completion, learners will be able to identify and describe stainless steels and related welding procedures.

- Objectives:**
1. Describe stainless steel.
 2. Identify types of stainless steel and their properties.
 3. Identify the AISI numbering system of stainless steel.
 4. Identify carbide precipitation and ways of overcoming this problem.
 5. Identify the major types of stainless steel filler materials and AWS specifications.
 6. Describe handling and storage of stainless steel electrodes and filler materials.
 7. Describe the handling procedures and preparation for welding stainless steel.

Topic C.	Nickel Alloys and Clad Steels	Weighting	
		4 Hours	1%

Outcome: Upon successful completion, learners will be able to describe nickel alloys, clad steels, and select filler metals.

- Objectives:**
1. Describe nickel and its alloys.
 2. Describe clad steels and their advantages.
 3. Select filler metals in AWS specifications.
 4. Describe preparation and welding procedures for clad steels.

Topic D.	Applied Metallurgy and Metal Identification	Weighting	
		6 Hours	2%

Outcome: Upon successful completion, learners will be able to identify space-lattice types and describe grain growth and grain structure associated with welding.

- Objectives:**
1. Identify space-lattice types in metals.
 2. Describe dendritic grain growth.
 3. Describe grain structure differences in metals.
 4. Identify changes in grain structure that result from welding.
 5. Interpret information supplied on mill test reports.

Topic E.	Non-Destructive Testing	Weighting	
		4 Hours	2%

Outcome: Upon successful completion, learners will be able to identify and describe NDT methods and their uses.

Topic E.	Non-Destructive Testing	Weighting	
		4 Hours	2%

- Objectives:**
1. Identify NDT.
 2. Describe visual inspection techniques.

Topic F.	Destructive Testing (DT)	Weighting	
		4 Hours	2%

Outcome: Upon successful completion, learners will be able to identify and describe DT methods.

- Objectives:**
1. Identify types of DT.
 2. Describe DT methods.

Section Three: Drafting, Drawings and Specifications



This section focuses on interpreting structural, pressure vessel, and piping drawings.

Topic A.	Structural Drawings	Weighting	
		16 Hours	6%

Outcome: Upon successful completion, learners will be able to analyze, identify, and interpret structural drawings.

- Objectives:**
1. Interpret abbreviations used on drawings.
 2. Identify site plans, benchmarks and orientation.
 3. Identify structural shapes and how they are specified.
 4. Identify types of structural drawings.
 5. Extract dimensions and other information from drawings.
 6. Interpret bill of materials for drawings.
 7. Interpret drawings.

Topic B.	Pressure Vessel Drawings	Weighting	
		16 Hours	7%

Outcome: Upon successful completion, learners will be able to analyze, identify, and interpret pressure vessel drawings.

Topic B.	Pressure Vessel Drawings	Weighting	
		16 Hours	7%

- Objectives:**
1. Identify external and internal vessel components.
 2. Identify material compositions as per code requirements on mill certification.
 3. Interpret vessel drawings.

Topic C.	Piping Drawings	Weighting	
		16 Hours	7%

Outcome: Upon successful completion, learners will be able to analyze, identify, and interpret piping drawings.

- Objectives:**
1. Describe the purpose of a spool sheet.
 2. Interpret symbols that represent individual components on a spool sheet.
 3. Interpret position and orientation of piping systems from an isometric drawing.
 4. Interpret component sizes in a piping system from a spool sheet.
 5. Develop isometric drawings from orthographic projections.
 6. Develop a material list for a piping system.
 7. Interpret pipe drawings.

Section Four: Shielded Metal Arc Welding (SMAW)



This section develops SMAW groove weld skills on mild steel plate and pipe.

Topic A.	SMAW on Mild Steel	Weighting	
		50 Hours	21%

Outcome: Upon successful completion, learners will be able to perform SMAW groove welds on mild steel plate.

- Objectives:**
1. Perform 2G, 3G, 45° overhead and 4G welds using E4310 (E6010) for the root pass and E4918 (E7018) for the fill and cap.
 2. Perform 4GF weld using E4918 (E7018) with backing according to CSA Standard W47.1.

Topic B.	SMAW on Mild Steel Pipe	Weighting	
		32 Hours	13%

Outcome: Upon successful completion, learners will be able to perform SMAW groove welds on mild steel pipe.

- Objectives:**
1. Demonstrate preparation, fit up, and tack weld pipe.
 2. Perform feathering on tack welds.
 3. Perform groove weld on 150 mm (6 in.) schedule 80 pipe in the 2G-5G positions using an E4310 (E6010) root and E4918 (E7018) fill and cap.
 4. Perform groove weld on pipe in the 5G position downhill.

Section Five: Gas Tungsten Arc Welding (GTAW)



This section develops GTAW skills on mild steel plate and pipe.

Topic A.	GTAW on Mild Steel Plate and Pipe	Weighting	
		40 Hours	17%

Outcome: Upon successful completion, learners will be able to prepare, fit-up, and weld using GTAW and SMAW on mild steel plate and pipe.

- Objectives:**
1. Demonstrate preparation and fit-up for GTAW.
 2. Perform 1G and 2G welds on mild steel plate.
 3. Perform 3G weld on mild steel plate using GTAW root and SMAW E4918 (E7018) fill and cap, uphill for all passes.
 4. Perform 2G, 5G and 6G welds on mild steel pipe.

Taxonomy Verb List

This is a list of commons verbs used to demonstrate the level and/or complexity of a given taxonomy. **It is only intended as a guide and is not meant to exclude additional verbs.**

Taxonomy I

Verbs	
Convert	Define
Describe	Explain
Identify	List
Recognize	State

Taxonomy II

Verbs	
Apply	Connect
Demonstrate	Draw
Extract	Locate
Maintain	Perform
Select	Splice
Tap	Terminate
Use	

Taxonomy III

Verbs	
Analyze	Calculate
Create	Determine
Develop	Interpret
Program	Solve
Troubleshoot	Verify



Apprenticeship
and Industry Training
tradesecrets.alberta.ca